Aarhus University

Department Of Mathematics

Introduction to Algebra 1

Handin 2

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Goofy ah Math Number 2

#Consistency

Determine $\gcd(175, 63)$.

Theorem 3.2 [1] Euclidean Algorithm. The algorithm is used as described in the book for $r_0 = 175$, $r_1 = 63$

k	r_k	q_k	r_{k+1}
0	175	-	63
1	63	2	49
2	49	1	14
3	14	3	$\underline{\underline{7}}$
4	7	2	0

Table 1: Euclidean Algorithm finding $\gcd(175, 63)$

Reiterating the conclusion in the table, $\gcd(175, 63) = 7$ since the remainder in the next step $k = 4$ is zero, then according Euclid's Algorithm the gcd is the r_{k+1} value from the $k - 1$ step, which is 7.

Determine the integers x and y so $7 = 175 \cdot x + 63 \cdot y$

Extended Euclidean algorithm - Bezouts identity: theorem 3.3 [1] The algorithm is used as is described in the book.

$$\begin{aligned}
x_0 &= 1, & y_0 &= 0, \\
x_1 &= 0, & y_1 &= 1, \\
x_k &= x_{k-2} - q_{k-1} \cdot x_{k-1}, & y_k &= y_{k-2} - q_{k-1} \cdot y_{k-1},
\end{aligned}$$

k	r_k	q_k	x_k	y_k
0	175	-	1	0
1	63	-	0	1
2	49	2	1	-2
3	14	1	-1	3
4	$\underline{\underline{7}}$	3	$\underline{\underline{4}}$	$\underline{\underline{-11}}$

Table 2: Extended Euclidean Algorithm finding integers x and y so $7 = 175 \cdot x + 63 \cdot y$

Double checking the result for $x = 4$ and $y = -11$

$$175 \cdot 4 - 63 \cdot 11 = 700 - 693 = 7$$

The values seems to match, so both x and y have been seriously determined and considered and look at and mathed REAL HARD...SERIOUSLY!

Convince God (Nicolaj) that there doesn't exist some integer r and s so $50 = 175 \cdot r + 63 \cdot s$

Corollary 3.5 [1]
For some numbers $a, b \in \mathbb{Z}$ that are not both zero, if $d = \gcd(a, b)$ then there is some integers r and s so that $c = ar + bs$ if and only if $d|c$.
This means there should exist some integer q for $d = 7$ and $c = 50$ such that $50 = 7q$. Sadly $\frac{50}{7} \notin \mathbb{Z}$ which means q doesn't exist and this follows that r and s don't exist either.

References

[1] Johan P. Hansen. *Tal og polynomier. Begreber, metoder, resultater, kodning og kryptografi*. Aarhus Universitetsforlag, 2018.